

EDUCATION

Bachelor of Engineering (Honours), Mechanical Engineering | UNSW Sydney **2025 - 2028 (expected)**
SACE | Pedare Christian College, Golden Grove SA | ATAR: 97.05 **2024**

TECHNICAL SKILLS

CAD and mechanical design: SolidWorks, Fusion 360; drone airframes, competition robots, antenna mounts and brackets
Programming: C++ for embedded and competition robotics; Python for automation, data analysis and trading APIs; MATLAB; TypeScript
Mechatronics and embedded: IMUs, encoders, colour and environmental sensors; flight controllers; pneumatics; Raspberry Pi and ADC acquisition

EXPERIENCE

Co-Founder and CFO | Project Umbrella Labs, UNSW **2025 - Present**
- Co-founded a student-led research and development lab running practical engineering projects across autonomous UAVs, neural interfaces and agricultural monitoring.
VEX Robotics Mentor (Casual) | Barker College, Hornsby NSW **2026 - Present**
- Coach VEX IQ teams through design, build and programming cycles using national-championship VRC competition experience.
Mechanical Lead | BlueSat UNSW CubeSat Program **Feb - Dec 2025**
- Led mechanical design of the ground-station antenna pointing mounts and bracketry, coordinating interfaces across electrical, software and mechanical subteams.
- Handed the mechanical package to the team while the ground station remained in development.
Front of House | Caffe Primo, Munno Para SA **Dec 2021 - Jan 2025**
- Handled orders, payments and bar service in a high-volume hospitality environment.

SELECTED PROJECTS

Cafe Primo Firl restaurant platform | Working restaurant **2026 - Present**
- Built and operate a production platform connecting customer ordering, Stripe checkout, live staff and admin dashboards, PostgreSQL, Socket.io updates and ESC/POS receipt printing.
ADA2M autonomous agricultural drone | Project Umbrella Labs, team of two **2025 - Present**
- Designed a custom carbon-fibre airframe around a SpeedyBee F405 flight controller, TBS Crossfire link, Jetson Nano and environmental sensor payload; currently in flight testing.
Automated day-trading platform | Personal project **2025 - Present**
- Building a modular Python system on the Interactive Brokers API with backtesting, position sizing, stop-loss logic and drawdown limits; no live performance claim.

LEADERSHIP, AWARDS AND SERVICE

Captain and Lead Programmer | VEX Robotics Team 41103A **2018 - 2024**
- Led a seven-person team to back-to-back Australian National Championships, two World Championship qualifications and the 2024 Worlds Sportsmanship Award in Dallas.
- Wrote autonomous and driver-control C++ using IMU, colour-sensor and encoder data; designed pneumatic intake, scoring and elevation mechanisms.
Volunteer | Little Sisters of the Poor and Order of Malta outreach **2024 - Present**
- Make weekly maintenance and companionship visits with elderly residents and distribute essential supplies through organised Sydney street outreach.

RECOGNITION

Australian Defence Force Future Innovators Award (2024)
Tournament Excellence, Australian National VEX Championship (2023)
Robotics work featured by ABC Radio Adelaide, Life FM and Glam Adelaide